

Novel Hyperbolic Curves Unique To Prime Numbers

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Abstract

We provide findings that all prime numbers $p \in P^+$ have a unique co-ordinate pair $pFrequency$ (pFr) and $pFactor$ (pFa) for $p - 1$, where $pFr \times pFa = p - 1$. As every prime number is unique, so is its $p - 1$. As every prime number is odd, so its $p - 1$ is even. This unique pair creates a ratio $rRatio$ that binds its prime to *only one discontinuous hyperbolic curve*, a new defining property of primes. The prime factor pFa determines and maps primes to its hyperbolic curve family. The basic formula $pRatio = pFactor \div pFrequency$ was recognized and reversed engineered from regression analysis equation $pRatio = pFa^2 \div (p - 1)$. While pRatios rarely share the same y-axis value, it is the co-ordinate pair that is unique. Yet primes of equal pRatios will not share the same pFactor or hyperbolic curve. Prime numbers $p[5,100000]$ were used. We present high precision examples, the study's methodology, means to determine the pFrequency of a prime, thus its pFactor and adjoining hyperbolic curve. We begin with a visual of increasing pRatios of this range.

1 Introduction

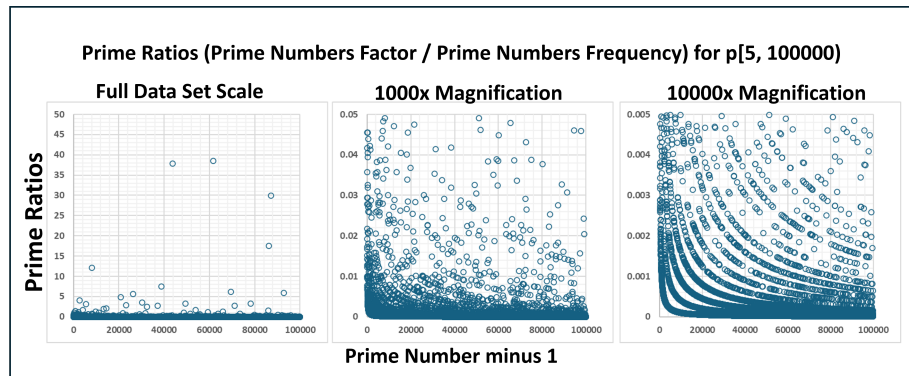


Figure 1: $p[5,100000]$ Full Data appears as noise. Patterns emerge at 1000x. 100000x shows ~ 20 curves. Its left lower curve is the basic inverse function: $y = x^{-1}$ but for the findings is expressed $y = pRatio = pFa^2 / (p - 1)$.

2 Equations, Table Generation, Methodology and Identification

2.1 Equations: Refer to Figure 2 for the subsections.

The driving functions come from another study around Mersenne Primes (not addressed here). Primes 2 and 3, represented in Row 3 and 4 respectively below, are not addressed due to their involvement in table generation.

1. Row 1 - Create a sequential top line row of $k \in \mathbb{N}_0$, starting at $k = 0$. It will connect iteration with primes as $k = p - 1$.
2. Row 2 - $f(k) = 2^k \rightarrow 2^{p-1}$
3. Row 3 - $g(x)_{Step1} : x - 1 \rightarrow g(f(k)) - 1 \rightarrow 2^{p-1} - 1$ (light blue line below, representing $p = 2$)
4. Row 4 - $g(y)_{Step2} : y \rightarrow g(x)_{Step1}/3$ (representing $p = 3$). A number is produced every other kth iteration (when k is even) [0, ,1, ,5, ,21, ,85, ,341, ,1365, ,5461, ,21845...]¹.
5. Generation of table: For Row 5 and forward $p[5,100000]$ - Check for whole number divisibility sequentially into Row 4 for each row of primes as divisors.

Figure 2: Formed composite table from $p[5...31]$

Observation: for prime p, beginning the k = p-1 iteration, p or p/q (mod g(f(2^k))) = 0 for some factor p-1/q (mod g(f(2^k)))																						
	k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
f(x)	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	
dR(g(f(x)) - Step 1 (-1))	0	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767	65535	131071	262143	524287	1048575	
- Step 2 (/3) >>> g(f(x))			1		5		21		85		341		1365		5461		21845		87381		349525	
5					1				17					273			4369				69905	
7							3							195					12483			
11											31										31775	
13														105								
17								5									1285					
19																			4599			
23																						
29																						
31											11										11275	

2.2 Methodology and Identification

- This produces a periodic divisible nature, e.g. row $p = 5$ observing (every 5 - 1 cells) implied an even frequency given for our A002450 series.
- This led to an uniqueness implication: $prime \rightarrow prime-1 \rightarrow (pFr, pFa) \rightarrow quotient \rightarrow pRatio$ of (pFr, pFa) .
- However at times this repeated at smaller rates of $f \in \mathbb{N}_{>1} : f \equiv 1 \pmod{p-1}$ and $1 < f < p-1$.²

¹comparable to <https://oeis.org/A002450>. For faster generation use $(n+1) = 4 * a(n) + 1$, with $a(0) = 0$ from <https://oeis.org/A002450>, Philippe Deléham, Feb 25 2004. NOTE: This list will be contiguous

² $pFa \neq 1$ and $pFa \neq p-1$ except for the special case of $pFr = 2$, not discussed here.

- ## 2.3 Identification and Goals

- Figure 3: Overlaying Figure 2: The arrows indicate periodic spacing for $pFr = p - 1$ and $pFa = 1$, like $p[5,7,11] \rightarrow pFr[4,6,10]$. But pFr may be less than $p - 1$, such as $p = 31$, $pFr = 11 \rightarrow pFa = 341/31 = 11$ at iteration $k = 10$, so $pFa = (31 - 1) / 10 = 3$. While linear is inefficient, it is the most thorough.

Table 1 - Observation: for prime p, at iteration k = p-1, $g(f(2^p)) \equiv 0 \pmod{p \text{ or } p - q}$, where $q \geq 3$																						
	k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
f(x)		1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576
dR(g(f(x)) - Step 1 (-))	0	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767	65535	131071	262143	524287	1048575	
dR(g(f(x)) - Step 2 (/3))	0		1	5		21		85		341		1365		5461		21845		87381		349525		69905
Prime : 5	0					1 5-1 cells				17 5-1 cells				273 5-1 cells				4369 5-1 cells				69905
7	0									3 7-1 cells				195 7-1 cells						12483 7-1 cells		
11	0												31 11-1 cells									31775
13	0												105									
17	0												5 17-1 / 2 cells						1285 17-1 / 2 cells			
19	0																			4599		
23	0																					
29	0																					
31	0													11 31-1 / 3 cells								11275

3 Various Examples

P	P-1	pRatio	pFr	pFa
1103	1102	0.32759	58	19
6043	6042	0.05975	318	19
7487	7486	0.04822	394	19
12503	12502	0.02888	658	19
15467	15466	0.02334	814	19
21851	21850	0.01652	1150	19
22573	22572	0.01599	1188	19
46703	46702	0.00773	2458	19
78509	78508	0.0046	4132	19
82499	82498	0.00438	4342	19
95989	95988	0.00376	5052	19
96331	96330	0.00375	5070	19

A scatter plot showing the relationship between pRatio (y-axis) and P-1 (x-axis). The data points are blue dots. A dashed blue line represents a power-law fit with the equation $y = 361x^{-1}$ and $R^2 = 1$. The x-axis ranges from 0 to 120,000, and the y-axis ranges from 0 to 0.35.

Figure 5: These four pFa curves show origination points of their hyperbolic trend lines and seem to lack some discernible correlation of when following pFa curves begin.

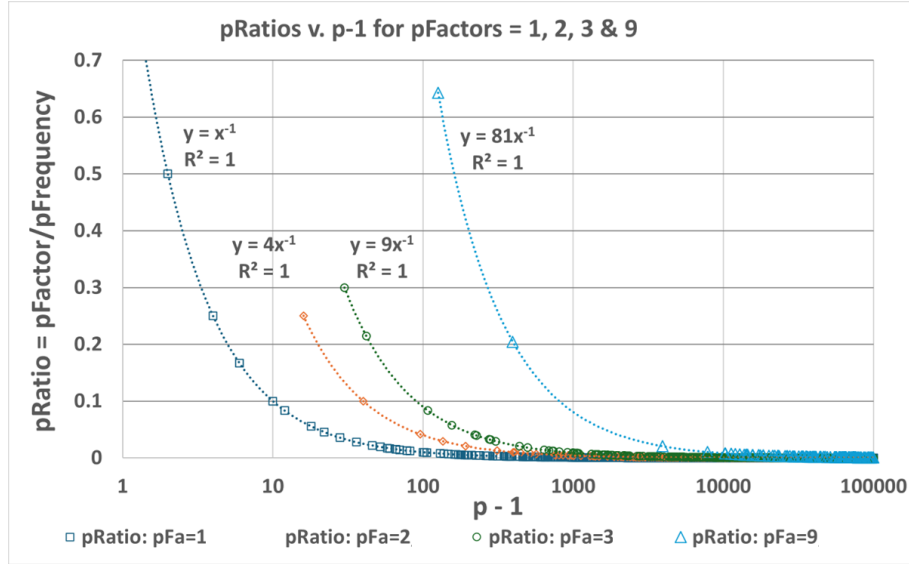
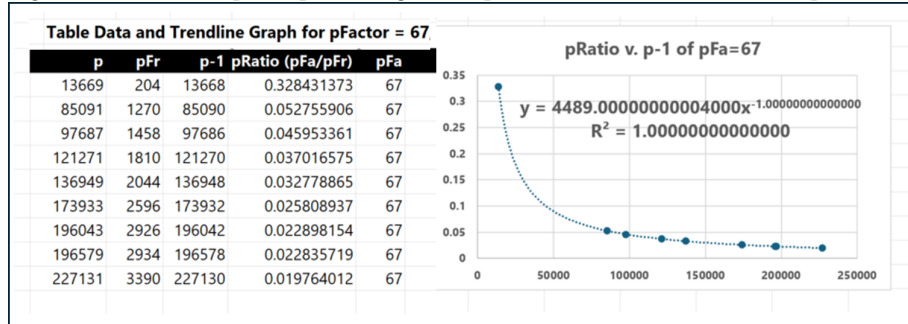


Figure 6: The same principle as Fig. 4 for pFa = 67 with more decimal precision.



4 Analysis and Observations

It is unclear why and when a new pFactor curve starts and the correlation to its first prime is number unknown.

- The first pFa, generates the largest pRatio for the curve, with the smallest $p - 1$ it will encounter, given thier inverse relationship.
- Co-ordinates will never be the same for any prime.
- Successive pFactors do not necessitate initial larger valued primes. See Table 1 below.
- Overall, triggering mechanisms for new pFa curves is unknown.

Figure 7: An earlier method to test precision (including other variables, not covered here).

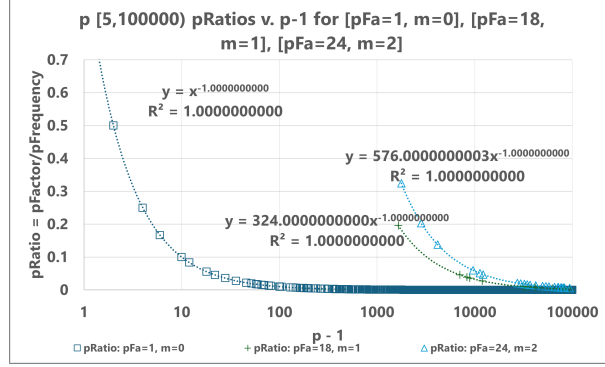


Figure 8: A group test on equally spaced pFa curves (here every 9th pFactor group). The results show comparatively sparse co-ordinates counts for succeeding pFa hyperbolas.

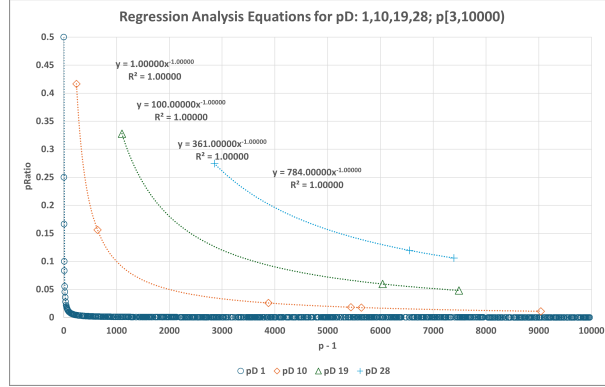


Table 1: First 30 pFactor Values and Their First Primes for p[5,100000)

pFa	1st Prime	pFa	1st Prime	pFa	1st Prime
1	5	11	331	21	2143
2	17	12	601	22	1321
3	31	13	4421	23	18539
4	73	14	673	24	1777
5	151	15	3061	25	2351
6	433	16	257	26	37441
7	631	17	1429	27	2971
8	337	18	1657	28	2857
9	127	19	1103	29	683
10	241	20	3121	30	17401

Figure 9: For a $p[5,100000)$ histogram, bin $[1,10]$ has 8841 of 9590 pFactors (92.2%), yet this 1 of 12 bins and a relatively small domain of primes.

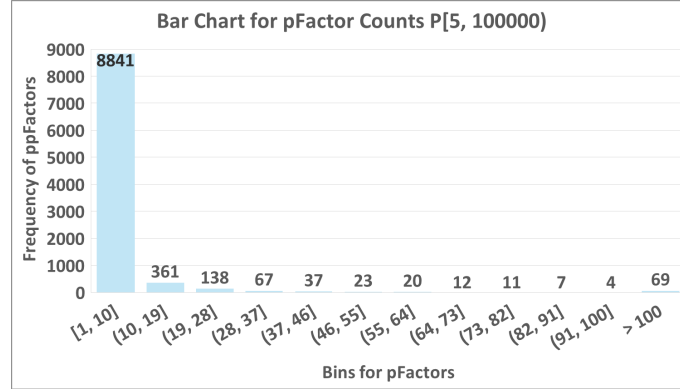


Table 2: Bin $[1,10]$ Quartile Data, pFa 1 has 5541 of 9590 or 56.4% pFactors yet there's notable variance within the bin.

pFactor	Q1	Q2	Q3	Q4	Mean	Std Dev	CoV	pFa Totals
1	1334	1363	1347	1367	1353	15.20	1.1%	5411
2	230	225	223	239	229	7.14	3.1%	917
3	252	252	226	231	240	13.72	5.7%	961
4	179	159	182	170	173	10.34	6.0%	690
5	78	49	75	43	61	17.82	29.1%	245
6	44	44	33	43	41	5.35	13.1%	164
7	35	28	35	43	35	6.13	17.4%	141
8	44	41	39	41	41	2.06	5.0%	165
9	21	31	25	31	27	4.90	18.1%	108
10	9	6	14	10	10	3.30	33.9%	39
QTotal	2226	2198	2199	2218	2210	13.96	0.6%	8841

5 Concluding Remarks

While the pFactors for $p[5,100000)$ seem fairly congregated, this is a relatively small magnitude of numbers. The co-ordinates (pRatio, $p - 1$) of the succeeding curves, do not have discernible patterns, like a prime's pFrequency and pFactor; however, understanding this could lead to an understanding of primes nature. Yet as intriguing beyond the triggering mechanisms for new pFa curves is when a point falls on a higher curve rather than the original, most dense: $pRatio = pFa^2/(p - 1)$, where $p = 2$, $pFr = 2$ and $pFa = 1$.